



## Research Paper

# Bayesian neural network for critical heat flux prediction: a unified approach with uncertainty quantification

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## ABSTRACT

Critical heat flux (CHF) determines the maximum transferable heat flux from the heated surface in thermal hydraulics systems. Beyond this limit, the local heat transfer coefficient (HTC) decreases dramatically leading to a boiling crisis or burnout of thermal equipment. To ensure the safety and safe operation as well as the maximum possible efficiency of thermal systems, accurate prediction of CHF is mandatory. Although hundreds of correlations have been proposed for subcooled and saturated flow boiling CHF, they are limited by certain operating conditions, including several empirical coefficients or based on the small amount of experimental data points. To develop a unified model, applicable for both types of CHF—DNB (departure from nucleate boiling, usually at low quality), and dryout (at high quality region), a machine learning (ML) approach has been adopted here. In this study, a Bayesian Neural Network (BNN) has been developed utilizing the NRC (Nuclear Regulatory Commission) CHF database to predict CHF as well as uncertainty in prediction. The NRC database consists of 24,579 CHF experimental data points collected over a span of 60 years. The model utilizes 80 % (19663) of data for training and 20 % (4916) for testing the model. Uncertainty in the prediction has been quantified by a 95 % confidence interval (CI). The prediction has been compared to a look-up-table (LUT) in terms of five statistical metrics- mean P/M (mean ratio of predicted and measured CHF), std P/M (standard deviation of mean prediction and measured CHF ratio), RMSPE (root-mean-square-percentage-error), MAPE (mean-absolute-percentage-error), and  $R^2$  (determination of coefficient). The developed BNN model predicts the test dataset with a mean P/M, std P/M, RMSPE, MAPE, and  $R^2$  of 1.007, 0.123, 12.3 %, 7.6 %, and 0.986, whereas these values for LUT are 1.029, 0.328, 32.9 %, 19.8 % and 0.941, respectively. Moreover, it captures around 94.8 % of test data within 95 % CI. Overall, BNN has superior performance over LUT for testing, the whole CHF database, and the unseen dataset (not used in training and testing) for CHF prediction although the model captured 64.7 % unseen data in 95 % CI. The predictions on the slice's dataset ensure that the model is capturing physics behind CHF perfectly. Moreover, the developed BNN model also has been evaluated to extrapolate (outside tube diameter, mass flux, and geometry) data points.

## 1. Introduction

Heat dissipation is the key requirement for the safe and normal operation of thermal systems that work on boiling and evaporation. Single-phase heat transfers are sufficient for the small amount of heat dissipation using its sensible heat. However, thermal components used in various industries—including steam generators, boilers, heat exchangers, high-powered electronic devices, and notably nuclear reactors—generate substantial amounts of heat that must be effectively removed from heated surfaces to prevent thermal failure. The necessity of high heat removal leads to a phase change heat transfer mechanism where the latent heat of working fluid is used to remove high heat flux

from the heated. Boiling or evaporation provides significantly higher heat transfer coefficients than single-phase flow [1]. However, in heat flux-controlled systems, heat transfer is limited by the Critical Heat Flux (CHF), the maximum allowable heat flux that can be safely removed from a heated surface. Exceeding CHF can lead to burnout or system failure, posing serious risks, especially in nuclear applications such as water-cooled nuclear reactors. In water-cooled nuclear reactors, exceeding CHF will cause a sharp rise in fuel rod clad temperature, leading to degradation of cladding material strength and possible oxidation, which can result in fuel cladding failure. This may, in turn, lead to core damage and potential radioactive release [2,3].

CHF can be experienced in both sub-cooled and saturated conditions of bulk liquids for a heated channel/ surface. Subcooled CHF occurs with

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**Nomenclature**

|                |                                                              |
|----------------|--------------------------------------------------------------|
| CHF            | Critical heat flux (kW/m <sup>2</sup> )                      |
| LUT            | Look-up-table [23]                                           |
| BNN            | Bayesian neural network                                      |
| RMSPE          | Root mean square percentage error                            |
| MAPE           | Mean absolute percentage error                               |
| R <sup>2</sup> | Coefficient of determination                                 |
| Mean P/M       | Mean ratio of predicted and measured CHF                     |
| Std P/M        | Standard deviation of mean prediction and measured CHF ratio |
| D              | Diameter (m) or dataset                                      |
| L              | Heated length (m)                                            |
| P              | Pressure (kPa)                                               |
| G              | Mass Flux (kg/m <sup>2</sup> s)                              |
| X              | Quality                                                      |
| y <sup>p</sup> | Predicted CHF in equations (11)–(15)                         |
| y <sup>m</sup> | Measured (experimental) CHF in equations (11)–(15)           |
| μ <sub>r</sub> | Mean of predicted/measured CHF values                        |
| μ <sub>m</sub> | Mean of measured CHF values                                  |

zero to low quality in the bubbly flow region which is called departure from nucleate boiling (DNB). On the other hand, saturated CHF is experienced in annular flow regions with higher quality which is known as dryout. At DNB an unstable vapor blanket is formed on the heated surface that restricts the liquid supply toward the heated surface resulting in the deterioration of Heat Transfer Coefficient (HTC) for the surface. This consequence causes a sudden increase in surface temperature. For the case of ‘dryout’ the thin liquid film on the heated surface completely evaporates and the HTC for the heated surface degrades sharply resulting in an abrupt increase in surface temperature, possibly burnout of thermal equipment. As CHF is a crucial thermal design parameter, an extensive study has been conducted for the accurate prediction of CHF in various working liquids, flow geometries, and operating conditions. Conventionally mechanistic models, empirical correlations, or LUT are used for CHF predictions [4,5]. The popular and well-accepted model includes dryout of liquid sublayer [6], bubble crowding [7], and boundary layer separation [8]. Additionally, a DNB prediction method using a two-fluid computational fluid dynamics (CFD) model was developed by Ouk and Sharma [9], while purely mechanistic models for predicting CHF in subcooled boiling have been proposed by Dong et al. [10], based on liquid sublayer depletion, and by Liu et al. [11], based on liquid sublayer dryout. Considering empirical correlations, Hall & Mudawar [12] conducted an extensive assessment of around 100 CHF correlations (subcooled) applicable to water flow in uniformly heated round tubes and proposed two CHF correlations in the dimensionless form which are widely acceptable for CHF prediction. The developed correlations are based on inlet conditions—recommended for water flow with uniform heat flux, and on local conditions that may apply to non-uniform axial heat flux. After evaluating 4 well-accepted correlations and 3 models, Celata et al. [13] established a correlation for CHF in subcooled boiling after modifying the experimental parameters in Tong [8]. Another correlation by modifying the experimental parameter has also been developed by Inasaka & Nariai [14] in subcooled flow boiling.

Although high HTC is obtained in the nucleate boiling region for flow boiling, HTC increases with the decrease of film thickness adjacent to the heated wall in the axial direction [15]. So, maximum HTC can be achieved in an annular flow region. However, in annular flow, the thin liquid film on the wall gradually diminishes due to the combined effects of increasing heat flux and liquid entrainment as droplets into the gas core. Eventually, the film is completely depleted from the heated surface, leading to dryout [16]. The flow pattern changes from annular mist

to mist flow with a sharp decrease in HTC at this condition. A simpler correlation proposed by Bowring [17] for the prediction of dryout in a uniformly heated round tube is one of the most accurate. This correlation only uses four optimized parameters as functions of pressure for CHF prediction. Katto [18–20] conducted an extensive study on CHF prediction in uniformly heated vertical tubes for saturated or dryout conditions. These works are also regarded as one of the most verified correlations. Proposed CHF correlations are based on vectorial dimensional analysis [18], the graphical method [19], and the concept of critical film thickness encountered in the annular flow regime [20]. The equation developed by Shah [21] is also in good agreement with experimental data collected from 62 independent sources covering 23 working fluids. Anglart et al. [22] developed a mechanistic CFD model for dryout and post-dryout temperature in diabatic annular flow, and recently an experimental work by Morse et al. [15] indicates that the disturbance waves in the annular region are the main contribution factor for dryout and rewetting process.

Besides the mechanistic model and empirical correlations, Look-Up-Table (LUT) by Groeneveld et al. [23] is also widely used for CHF prediction. Note that the word ‘LUT’ indicates this reference only in the whole article, including tables and figures. This is a normalized databank of an 8 mm vertical water-cooled tube that predicts CHF as a function of inlet mass flux, system pressure, and thermodynamic quality. CHF can be calculated on two conditions— constant inlet and constant outlet conditions. First, CHF is obtained for an 8 mm diameter tube, then the original tube diameter factor is multiplied to get real CHF. LUT table is relatively easy to use as it includes only 3 input parameters, the data set covers a wide range of operating conditions, and even no iteration is required for CHF prediction. Although three methods are available for CHF prediction, empirical correlations are always preferred for CHF prediction in thermal system design [12].

Although hundreds of correlations have been developed for CHF prediction, they are typically limited to specific operating conditions, working fluids, or flow channel geometries. As a result, significant deviations may occur when these correlations are applied outside their intended domains. Moreover, many of these empirical correlations are validated using relatively small experimental datasets, making them difficult to generalize for broader applications. Even when some correlations demonstrate reasonable accuracy, they involve many empirical constants and conditions. Similarly, certain mechanistic models incorporate parameters derived from experimental data, limiting their predictive versatility. Therefore, there is a clear need for unified models that are applicable across a wide range of operating conditions and validated against extensive experimental data. Recently, the U.S. Nuclear Regulatory Commission (NRC) released a CHF databank [24] comprising 24,579 experimental data points collected over a span of 60 years. This comprehensive databank includes CHF measurements for uniformly heated circular tubes using water as the working fluid. The objective of this work is to develop a machine learning model that can predict CHF with greater accuracy and reliability than existing methods, while also incorporating uncertainty quantification.

Machine Learning (ML) has demonstrated superior predictive capability over mechanistic and empirical correlations, making it a valuable tool in thermal hydraulics, including nuclear reactor applications [25]. Su et al. [26] developed an artificial neural network (ANN) for CHF prediction under different operating conditions (inlet, local, and outlet), achieving  $\pm 10\%$  accuracy for local conditions using pressure, mass flow rate, and equilibrium quality as inputs. Zhang et al. [27] employed a Deep Feed Forward Neural Network (DFNN) and Random Forest (RF) using six input parameters (considering both equivalent and heated tube diameter, and heated length along with the parameter of Su et al. [26]) for DNB type CHF prediction, with improved accuracy when inlet sub-cooling replaced quality. A DNN was also used to predict reactor design parameters, showing good agreement with RELAP5/SCDAPSIM [28]. An extensive study [5] evaluated various ML algorithms (ANN, KNN, AdaBoost, XGBoost, RF) for CHF prediction from 26 data sources, where

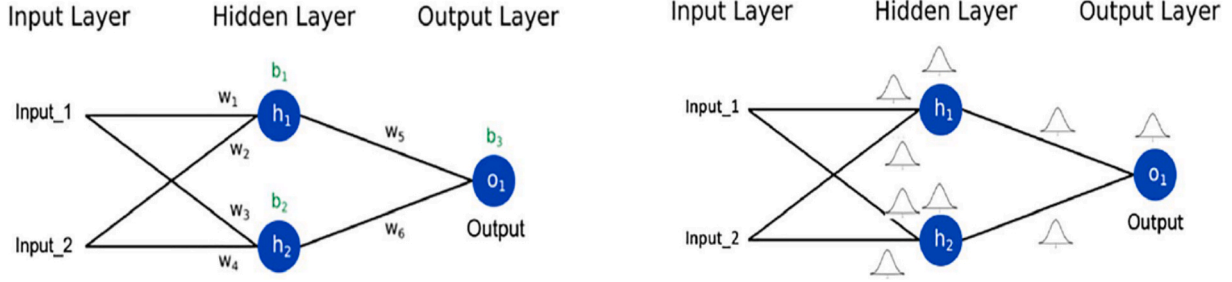


Fig. 1. Sample of standard neural network (left) and Bayesian neural network (right) [41].

ANN performed best (MAE: 8.85 %,  $R^2$ : 0.988). Modified models such as Genetic Neural Networks [29] and Cascade Feed Forward Neural Networks (CFNN) [30,31] further improved CHF prediction in boiling channels. ANN has also been applied to identify key dimensionless groups [32], CHF temperature [33], and conduct sensitivity analysis [34] in CHF prediction. Support Vector Regression (SVR) has shown strong potential for CHF prediction. Jiang et al. [4] demonstrated that SVR performs comparably to Radial Basis Function (RBF) neural networks across different inlet conditions in vertical round tubes. Cai [35] found SVR more accurate than the BP model for CHF prediction in a thermosiphon. Another study [36] used SVR with mass flux, pressure, and equilibrium quality as inputs, identifying mass flux and pressure as the most influential parameters for accurate CHF prediction. Recently, a physics-guided neural network has also been used for CHF prediction in narrow rectangular channels under different operating conditions [37]. It was reported that prediction error decreased by 10–5 % compared to empirical correlations. In addition, machine learning models have been used to predict CHF in solid fuels [38] and in pool boiling by images [39].

Numerous Machine Learning (ML) models have been proposed for predicting Critical Heat Flux (CHF) in two-phase flow. However, most of these models are deterministic, providing only point estimates without quantifying uncertainty. Although Soibam et al. [40] performed an uncertainty analysis using Parametric Gaussian Process Regression (pGPR), the study was limited to a small dataset (513 samples) and did not distinguish between aleatoric and epistemic uncertainties. Uncertainty quantification (UQ) is a crucial component of ML modeling, as it enhances model reliability and applicability across diverse conditions. The present study employs a Bayesian Neural Network (BNN) for CHF prediction, incorporating both epistemic uncertainty—stemming from limited knowledge about the model or process—and aleatoric uncertainty, which arises from inherent variability in the data. Unlike conventional ML models that yield only deterministic outputs, BNNs provide a distribution of predictions, enabling both mean estimates and associated uncertainties. The predicted uncertainties are evaluated using Confidence Interval (CI) scores. The model's predictive performance is assessed on both test and unseen datasets, with a comparison made against lookup table (LUT) predictions [23]. Error metrics such as mean and standard deviation of P/M, Root Mean Square Percentage Error (RMSPE), Mean Absolute Percentage Error (MAPE), and the coefficient of determination ( $R^2$ ) are used for evaluation. Finally, the model's extrapolation capability is analyzed by testing its performance on data with unseen geometries.

## 2. Methodology

### 2.1. Bayesian inference

Probabilistic Bayesian Neural Network (BNN) is an extension of conventional neural networks that mainly focuses on the prediction of both model uncertainty (epistemic) and data (aleatoric) uncertainties. In the standard neural network, the model learns a single weight for a

deterministic prediction. However, in BNN model learns a distribution (often Gaussian) of weights instead of a particular weight. A sample of standard neural networks and Bayesian neural networks is illustrated in Fig. 1 [41]. The uncertainties are captured by the distribution nature of weights. BNN uses Bayesian inference for posterior distribution. According to Bayes' theorem, posterior distribution is related to prior and likelihood by following,

$$p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)} = \frac{p(D|\theta)p(\theta)}{\int p(D|\theta)p(\theta)d\theta} \quad (1)$$

where,  $p(\theta)$  is the prior distribution (prior belief) of the parameter (weights/bias). This is the initial belief of parameter distribution before seeing data.  $p(D|\theta)$  likelihood, probability of observed dataset for given parameters.  $p(\theta|D)$  is the posterior distribution (probability of parameter given the dataset). This is the updated belief of a parameter after seeing the new data.  $p(D)$  is the marginal likelihood (evidence)

For a supervised regression problem, the dataset can be written as,  $D = \{(x_i, y_i)\}_{i=1}^N$ , where  $x_i$  is the input feature and  $y_i$  is the corresponding output. The likelihood of observing  $y_i$ , given the input  $x_i$  and parameter  $\theta$  is  $p(y_i|x_i, \theta)$ . For all the data, the likelihood can be written as,

$$p(D|\theta) = \prod_{i=1}^N p(y_i|x_i, \theta) \quad (2)$$

However, the posterior distribution can't be determined directly due to the intractable nature of the evidence. To overcome this issue, the variational inference or Markov chain Monte Carlo method is used to estimate posterior distribution. TensorFlow Probability offers a DenseFlipout layer for the variational inference method.

### 2.2. Variational inference

A simpler distribution,  $q_\phi(\theta)$  parameterized by  $\phi$ , is used to approximate the true distribution,  $p(\theta|D)$ . The difference between the true distribution and approximate distribution is evaluated by the term Kullback-Leibler (KL) divergence. The KL divergence between approximate posterior and true posterior is defined as,

$$KL(q_\phi(\theta)||p(\theta|D)) = \int q_\phi(\theta) \log \frac{q_\phi(\theta)}{p(\theta|D)} d\theta \quad (3)$$

To replace the intractable posterior with tractable variational posterior approximation, Bayesian inference is applied in the above equations. After applying Bayesian inference in Eq. (3), Eq. (4) is obtained,

$$KL(q_\phi(\theta)||p(\theta|D)) = \int q_\phi(\theta) \log \frac{q_\phi(\theta)p(D)}{p(D|\theta)p(\theta)} d\theta \quad (4)$$

After splitting and separating the integral, the KL divergence term becomes,

**Table 1**

Data used for training and testing the BNN model (OECD database).

|     | Diameter<br>(m) | Heated<br>Length<br>(m) | Pressure<br>(kPa) | Mass<br>Flux<br>(kg/<br>m <sup>2</sup> s) | Quality | CHF<br>(kW/m <sup>2</sup> ) |
|-----|-----------------|-------------------------|-------------------|-------------------------------------------|---------|-----------------------------|
| Max | 0.016           | 20                      | 20,000            | 7964                                      | 0.999   | 16339.3                     |
| Min | 0.002           | 0.05                    | 100               | 8.2                                       | -0.497  | 50                          |

**Table 2**

Search space and best hyperparameters from Bayesian Optimization.

| Hyperparameter          | Search Spach | Best Hyperparameter  |
|-------------------------|--------------|----------------------|
| Number of neurons/units | 50–256       | 5–199–199–99–66–49–2 |
| Number of layers        | 3–8          | 5                    |
| Learning rate           | 0.0001–0.001 | 0.000887             |
| Batch size              | 128–712      | 463                  |
| Optimizer               | Adam         | Adam                 |
| Activation              | [RELU, ELU]  | RELU                 |
| Epochs                  | 1000–2000    | 1741                 |

$$KL(q_{\phi}(\theta)||p(\theta|D)) = \log p(D) - \int q_{\phi}(\theta) \log p(D|\theta) d\theta + KL(q_{\phi}(\theta)||p(\theta))$$

$$= \log p(D) - E_{q_{\phi}(\theta)} \log p(D|\theta) + KL(q_{\phi}(\theta)||p(\theta))$$

The final form becomes,

$$\log p(D) = ELBO + KL(q_{\phi}(\theta)||p(\theta|D)) \quad (5)$$

where, ELBO (evidence lower bound) is written as,

$$ELBO = E_{q_{\phi}(\theta)} \log p(D|\theta) - KL(q_{\phi}(\theta)||p(\theta))$$

The first term at the right side of the ELBO is the expected loglikelihood of the data given the parameters under the variational posterior,  $q_{\phi}(\theta)$ . The second term is the KL divergence between the approximate posterior,  $q_{\phi}(\theta)$ , and the prior,  $p(\theta)$ . As the log marginal likelihood is a constant i.e. doesn't depend on  $q_{\phi}(\theta)$ , minimizing KL divergence in Eq. (5) maximizes ELBO. Minimizing KL divergence means reducing the difference between true posterior and variational (approximate) posterior distribution. On another side, maximizing ELBO ensures variational posterior  $q_{\phi}(\theta)$  close to prior  $p(\theta)$ , and a decrease of KL divergence term prevents  $q_{\phi}(\theta)$  from overfitting also. The expected loglikelihood term defines how well the variational posterior explains the data for given parameters.

Once the posterior is determined, it can be used to assess the model

fit to the data or make a new prediction (new observation,  $\hat{y}$ ) on the new input,  $\hat{x}$ . Then the posterior predictive distribution becomes,

$$p(\hat{y}|\hat{x}, D) = \int p(\hat{y}|\hat{x}, \theta) p(\theta|D) d\theta \quad (6)$$

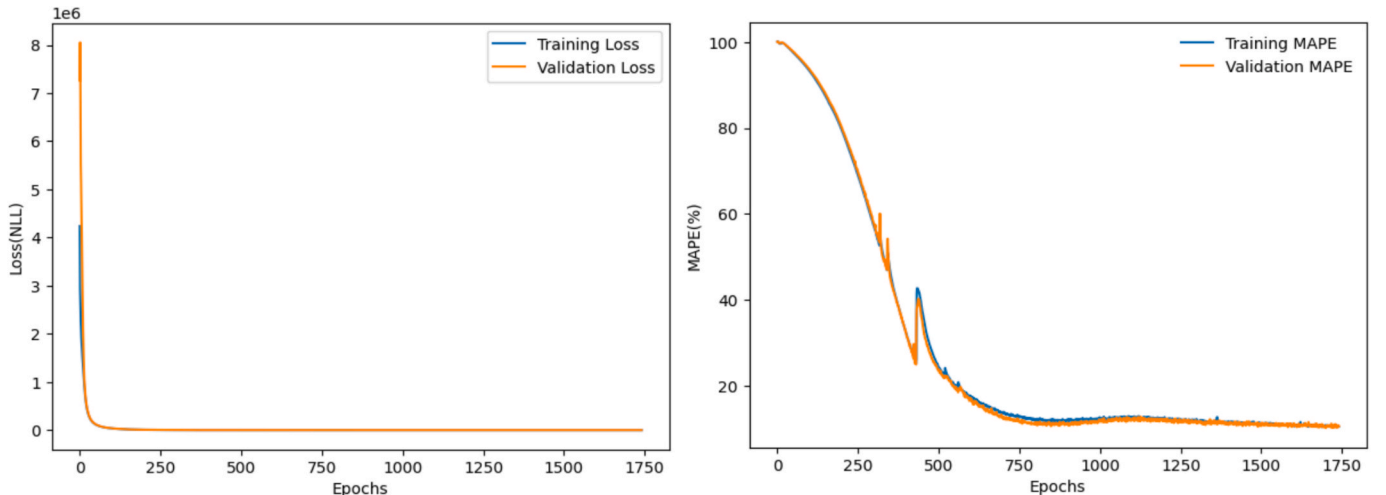
As weights are distributions unlike standard neural networks where it's a fixed value, uncertainty can be estimated in predictions. Both the aleatoric (data uncertainty) and epistemic uncertainty (model uncertainty) are determined from the right-side integral of the Eq. (6). However, the above integration term is also intractable. So, in the traditional Bayesian method the integral is approximated using Monte Carlo sampling,

$$p(\hat{y}|\hat{x}, D) \approx \frac{1}{M} \sum_{m=1}^M p(\hat{y}|\hat{x}, \theta^m) \quad (7)$$

here,  $\theta^{(m)}$  are the samples drawn from the posterior distribution,  $p(\theta|D)$ , and  $M$  is the number of Monte Carlo samples.

### 2.3. Bayesian inference in Tensorflow probability

TensorFlow Probability (TFP) is a library in TensorFlow designed for probabilistic reasoning and analysis. TFP provides tools and layers to implement Bayesian inference i.e. approximating posterior distribution. In TFP two methods-variational inference (VI) and Markov Chain Monte Carlo (MCMC) are used for the estimation of posterior distribution. For the current study, the VI method has been utilized for posterior estimation. TFP offers a DenseFlipout layer for variational inference internally using `tfp.distributions.kl_divergence` and `tfp.vi` [42]. VI approximates the posterior by minimizing the KL divergence between the approximated posterior and true posterior. In Keras API, KL divergence is automatically added to the loss function. For probabilistic modeling, negative loglikelihood is considered as a loss function. Probabilistic distribution (normal, Bernouli etc.) is defined by `tfp.distribution` module. TFP leverages VI to directly parameterize the weight distribution. As a result, the mean and variance of the posterior predictive distribution are obtained directly as output without the need for explicit Monte Carlo sampling like in the traditional Bayesian inference technique. Aleatoric uncertainty and mean prediction can be determined by a single pass of data into the model. However, for epistemic uncertainty, several passes are required, and uncertainty is calculated based on the standard deviation of mean prediction.

**Fig. 2.** Training and validation loss & MAPE with epochs.



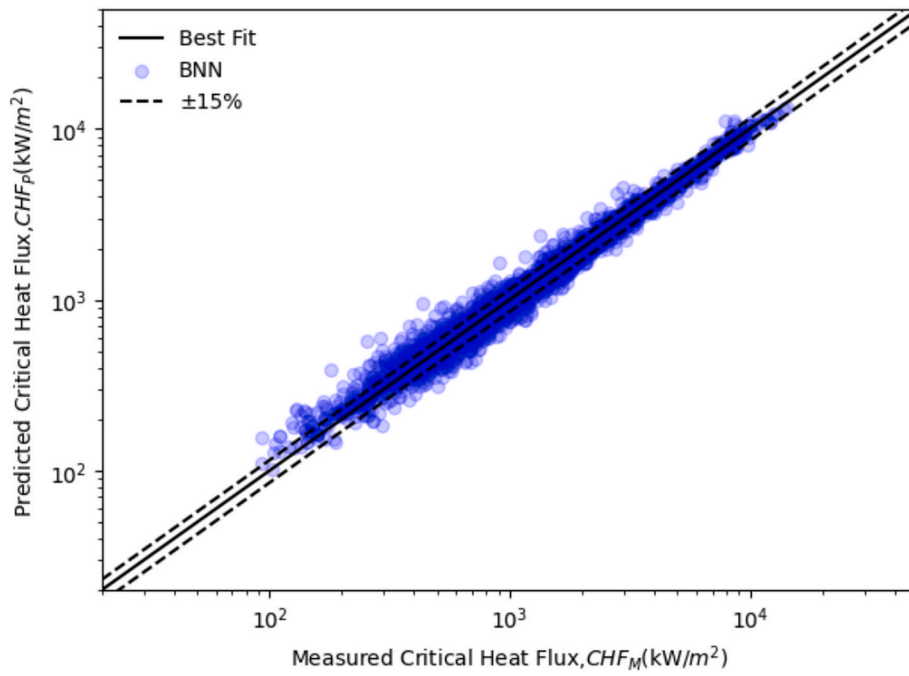


Fig. 3. Comparison of measured and predicted data (test) by BNN model.

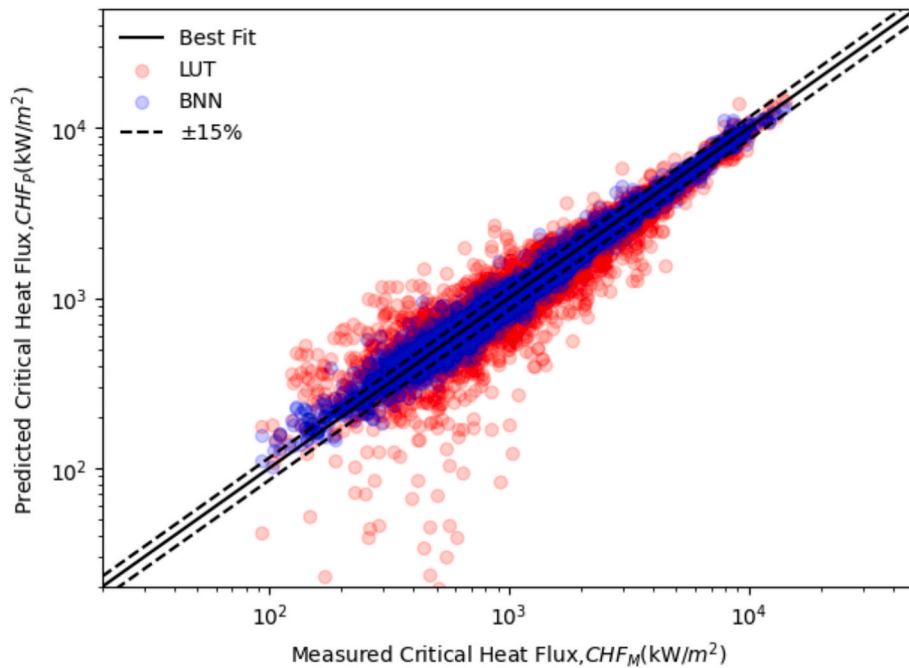


Fig. 4. Comparison of measured and predicted data (test) by BNN & LUT.

#### 2.4. Data source

The CHF database published by the US Nuclear Regulatory Commission (NRC) has been taken from the OECD Nuclear Energy Agency GitLab [24]. This database contains 24,579 CHF data points from the experiment conducted in the last 60 years. The data points include the measured parameters of the pressure, mass flux, inlet temperature, CHF, geometrical parameters of tube diameter and heated length, along with calculated parameters of outlet equilibrium quality and inlet enthalpy. For the current model development, the following 5 parameters have been used as input parameters, and their ranges are given in Table 1

below.

#### 2.5. Input parameters for BNN

CHF models are mainly a function of flow conditions (pressure, inlet temperature, mass flux, inlet subcooling, inlet or outlet quality) and channel geometry (tube diameter, heated length, or subchannel geometry for fuel assembly). For the development of CHF correlations, usually two types of conditions— inlet and outlet— are considered [12]. Inlet conditions (also called upstream conditions) are based on inlet mass flux ( $G$ ), pressure ( $P$ ), inlet enthalpy ( $h_i$ ) (or inlet subcooling,  $\Delta h_i$  or inlet

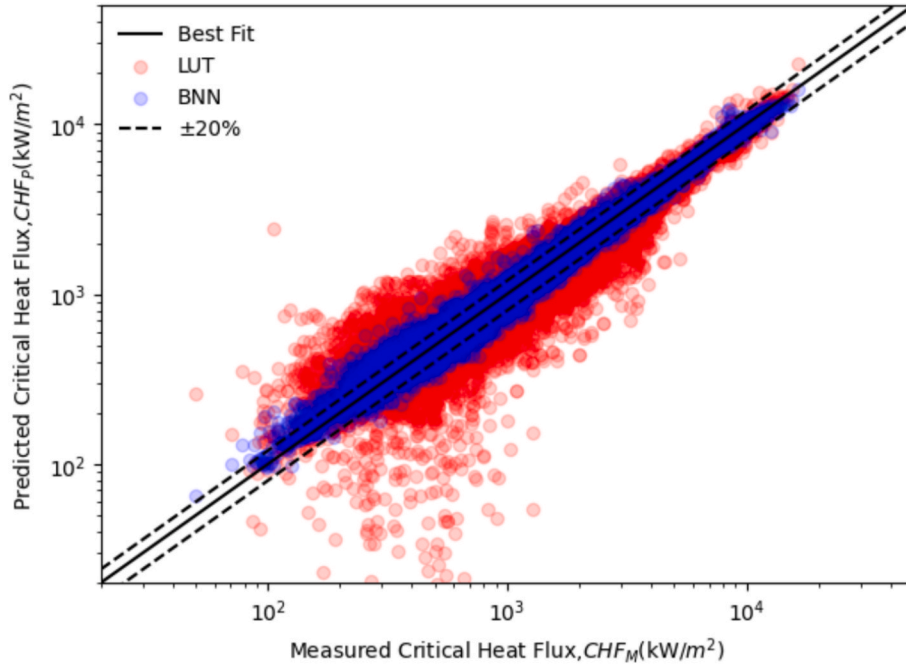


Fig. 5. Comparison of measured and predicted data (complete CHF database) by BNN & LUT.

Table 3

Performance evaluation of BNN & LUT on test and all datasets.

|      | Models | Mean P/M | Std P/M | RMSPE (%) | MAPE(%) | R <sup>2</sup> |
|------|--------|----------|---------|-----------|---------|----------------|
| Test | BNN    | 1.007    | 0.123   | 12.3      | 7.6     | 0.986          |
|      | LUT    | 1.029    | 0.328   | 32.9      | 19.8    | 0.941          |
| All  | BNN    | 1.003    | 0.106   | 10.6      | 6.5     | 0.989          |
|      | LUT    | 1.032    | 0.361   | 36.3      | 19.77   | 0.941          |

quality,  $x_i$ ) along with geometrical parameters such as heated length, tube diameter, etc.

$$q_{CHF} = f(D, L, G, P, \Delta h_i) \quad (8)$$

The second approach is based on outlet conditions (local conditions). Here, CHF is a function of outlet enthalpy ( $h_o$ ) (or quality,  $x_o$  or outlet subcooling,  $\Delta h_o$ ).

$$q_{CHF} = f(D, G, P, \Delta h_o) \quad (9)$$

For the current BNN, outlet conditions have been considered and, tube diameter, heated length, inlet mass flux, pressure, and outlet equilibrium quality have been used as input parameters to predict CHF as an output with both types of uncertainties.

## 2.6. Hyperparameter optimization (HPO)

The neural network normally includes many hyperparameters—the number of hidden layers, units or neurons in the layer, activation function, optimizer, learning rate, epochs, and batch size. The performance of the network is dependent on the selection of these hyperparameters. When the model is trained with the best parameters, it produces less error. Selecting hyperparameters manually is cumbersome and the process needs much trial and error. Normally, automated hyperparameter optimization methods are used for selecting the best parameters including Bayesian optimization (BO) [43], Random search [44], Grid search [45] methods, etc. These methods search hyperparameters in a given space by lowering the prediction error (minimizing evaluation metrics, normally mean squared error, mean absolute percentage error, mean absolute error, etc.). Although Random search

and Grid search are relatively easy to apply, BO is used where finding hyperparameters is relatively difficult. BO is an efficient technique to find hyperparameters for complex problems. BO uses a surrogate model, usually the Gaussian Process (GP) [46], to minimize an objective function. The Gaussian Process is a probabilistic model that predicts a mean value with uncertainty. BO technique learns the relation between hyperparameters and objective function, and based on the true value of objective function and using an acquisition function it selects new parameters set to reduce objective function and uncertainty in prediction. The surrogate model is updated based on the objective function evaluation (each new observation). So, a sufficient number of observations ( $n$  calls) are required to update the surrogate model to approximate the parameters for better accuracy of the objective function. Scikit-optimize (known as skopt), a Python library, is specially designed for Bayesian optimization. It uses a function, `gp_minimize`, to minimize the objection function for a defined set of parameters (space). This function uses GP to model the function and Expected Improvement (EI) as an acquisition function to select a new set of parameters (sample points) for evaluation (minimizing objective function). The hyperparameters, search space, and best hyperparameters after using BO are given in Table 2.

## 2.7. Model training

The BNN model was trained on Google Colab Jupyter Notebook. The best hyperparameters were obtained from the Bayesian HPO method discussed earlier. Then the model was trained again with the same training and test ratio (80 % of data for training and 20 % for testing). The HPO and final model used Negative-Loglikelihood (NLL) as the cost function and Mean-Absolute-Percentages Error (MAPE) as the evaluation metrics during training and validation. MinMaxScaler was utilized for scaling all data points used in this study. For a normal distribution as an output from the model, the log-probability is written as,

$$\log p(y|x, \theta) = -\frac{1}{2} \left( \log(2\pi\hat{\sigma}^2) + \frac{(y - \hat{\mu})^2}{\hat{\sigma}^2} \right) \quad (10)$$

And the negative-loglikelihood becomes,  $NLL = -\frac{1}{N} \sum_{i=1}^N \log(y_i|x_i, \theta)$  where,  $y$  is the observed (measured) values,  $x$  is the input, and  $\hat{\mu}$ ,  $\hat{\sigma}^2$  are predicted mean and variance respectively.

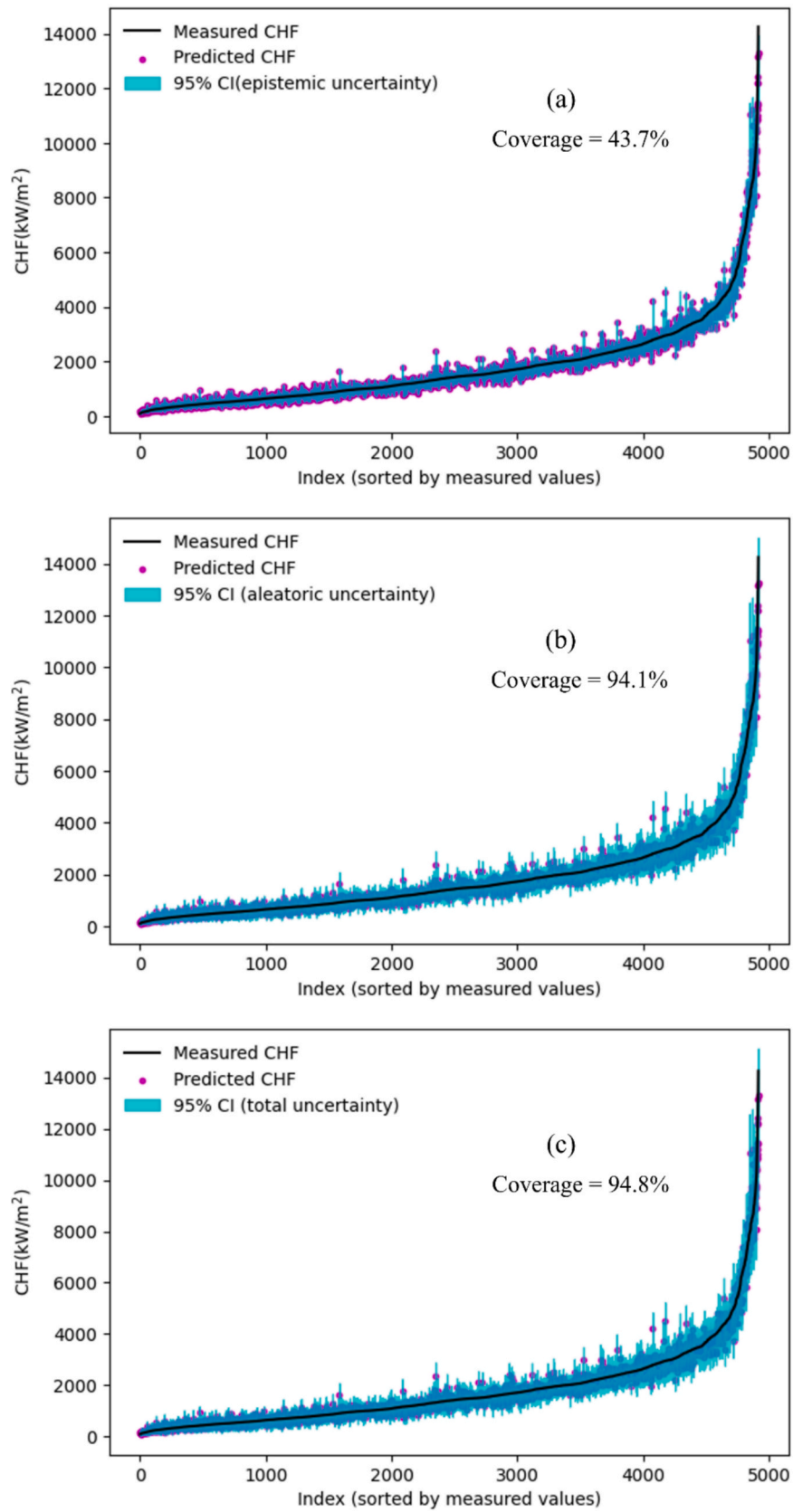


Fig. 6. Uncertainty quantification of test data with 95% CI. a) epistemic uncertainty. b) aleatoric uncertainty. c) total uncertainty.

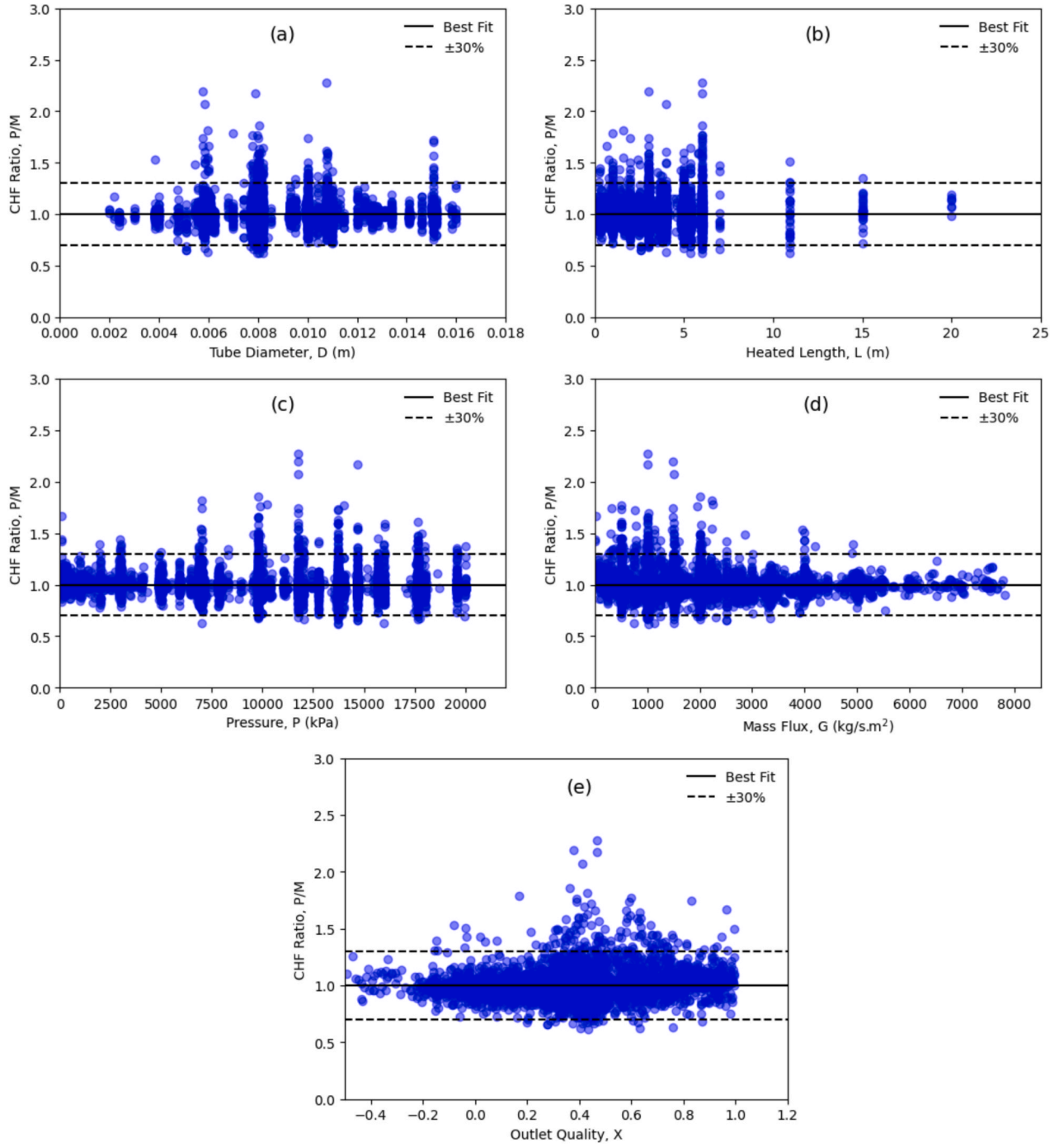


Fig. 7. Prediction accuracy on test data for input parameters. a) for diameter. b) for heated length. c) pressure. d) for mass flux, and e) for quality.

Several evaluation metrics have been considered for performance evaluation. They are as follows-Mean ratio or Bias ratio,

$$\text{Mean } P/M = \frac{1}{N} \sum_{i=1}^N \frac{y_i^p}{y_i^m} \quad (11)$$

The mean  $P/M$  metric determines whether the model tends to overfit or underfit on average. For the best performance, this metric should have a value of around 1. A value greater than 1 indicates over-prediction, whereas less than 1 indicates underprediction. It's a simple way to check model bias but doesn't give info about variability or spread.

Standard deviation of  $P/M$  (Std  $P/M$ ),

$$\sigma\left(\frac{y_i^p}{y_i^m}\right) = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(\frac{y_i^p}{y_i^m} - \mu_r\right)^2} \quad (12)$$

This metric quantifies the variability or spread of the ratio between predicted and measured CHF. A lower value is expected for better consistency in prediction. A higher value indicates high variability—high inconsistency. Here,  $\mu_r$  is the mean of predicted/measured values.

Mean-Absolute-Percentage-Error (MAPE):

$$\text{MAPE} = \frac{1}{N} \sum_{i=1}^N \left| \frac{y_i^p - y_i^m}{y_i^m} \right| \times 100 (\%) \quad (13)$$



**Table 4**  
Slices for BNN and LUT.

| Slices | Diameter (m) | Heated Length (m) | Pressure (kPa) | Mass Flux (kg/m <sup>2</sup> s) | Quality      |
|--------|--------------|-------------------|----------------|---------------------------------|--------------|
| 1      | 0.002–0.016  | 6                 | 14701.25       | 998.5                           | 0.3908       |
| 2      | 0.002–0.016  | 6                 | 9806.667       | 1003.33                         | 0.5291       |
| 3      | 0.008007     | 0.05–20           | 9806           | 1000                            | 0.5867       |
| 4      | 0.00811      | 0.05–20           | 2008.571       | 752.1714                        | 0.7562       |
| 5      | 0.008        | 0.99777           | 100–20000      | 2005.667                        | 0.1401       |
| 6      | 0.0134       | 3.658             | 100–20000      | 2040.197                        | 0.3777       |
| 7      | 0.008        | 1.57              | 12,750         | 8.2–7964                        | 0.1441       |
| 8      | 0.01         | 4.966             | 16,000         | 8.2–7964                        | 0.3433       |
| 9      | 0.00814      | 1.9428            | 9831.786       | 1519.511                        | −0.497–0.999 |
| 10     | 0.008004     | 0.99703           | 17,650         | 2002.722                        | −0.497–0.999 |

MAPE determines the absolute prediction error relative to measured values in percentage. A lower percentage error indicates high prediction accuracy. However, this metric is sensitive to lower measure values.

Root-Mean-Square-Percentage-Error (RMSPE):

$$RMSPE = \sqrt{\frac{1}{N} \sum_{i=1}^N \left( \frac{y_i^p - y_i^m}{y_i^m} \right)^2} \times 100 (\%) \quad (14)$$

This metric measures the relative error of the model's prediction in percentage. It is useful in performance evaluation, especially when dealing with datasets where values vary significantly in magnitude. A lower value of RMSPE is expected for higher accuracy.

Coefficient of Determination ( $R^2$ ):

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i^p - y_i^m)^2}{\sum_{i=1}^N (y_i^m - \mu_m)^2} \quad (15)$$

where,  $y^p$  is the predicted value (CHF) from BNN or LUT, and  $y^m$  is the

measured or experimental CHF for all cases, and  $\mu_m$  is the mean of measured values.

Lastly, the statistical measure  $R^2$  is used to assess the fraction of the variance (variation in the dependent variable) that can be explained by the developed model. A value close to 1 is desired for higher accuracy.

### 3. Results & discussion

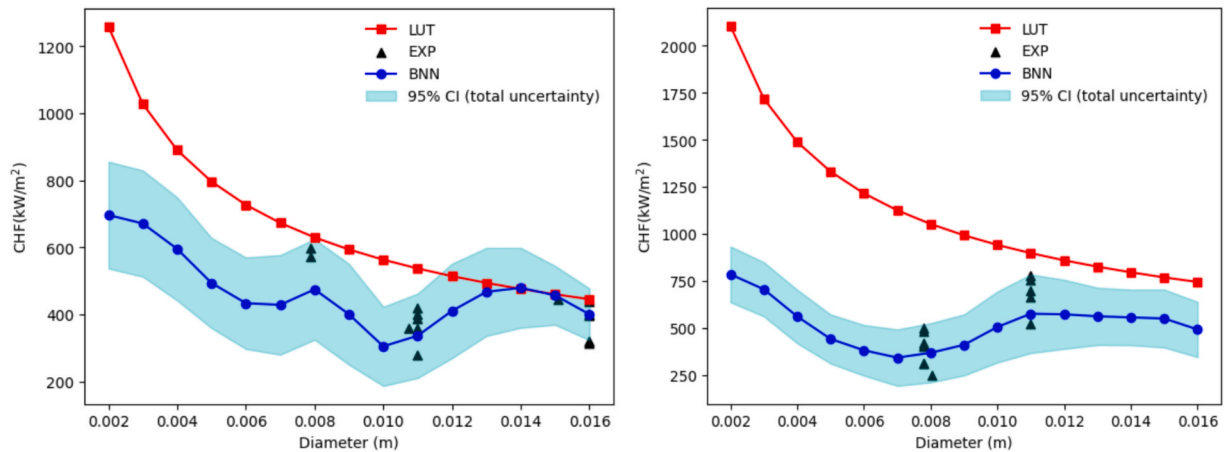
The training and validation loss (NLL) and metrics (MAPE) after training with the best HPO are demonstrated in Fig. 2. Although the loss was high initially, they reduced to a lower value rapidly. Evaluation metrics also showed the same trend. The loss and evaluation metrics didn't show any significant underfitting or overfitting as they were almost superimposed on each other. After getting the final BNN model, it was used for a prediction on the test, slice, and extrapolated dataset to depict its prediction accuracy with a comparison to LUT. The description of the sliced dataset is discussed in section 3.2, while the description of the extrapolated dataset is provided in section 3.4. The performance of the BNN model is discussed in the following sections.

#### 3.1. Prediction of test data with uncertainty

The developed BNN model is tested on a 20 % (4916 data points) test dataset to evaluate its prediction performance with  $\pm 15$  % error bound. The prediction on the test dataset is shown in Fig. 3. Another prediction by LUT on the same test dataset is also shown in Fig. 4. Furthermore, a comparison of the total 24,579 data points by the BNN and LUT is shown in Fig. 5. The improvement of CHF prediction accuracy of the BNN over LUT is depicted in Table 3 employing 5 evaluation metrics discussed earlier. Although the mean P/M is almost the same for BNN and LUT on the test dataset, the values for all other metrics vary significantly. LUT shows RMSPE, MAPE, and std P/M values of 32.9 %, 19.8 %, and 0.328 respectively, which reduces considerably for BNN, with only 12.3 %, 7.6

**Table 5**  
Range of experimental data for slices.

| Slices | Diameter (m)    | Heated Length (m) | Pressure (kPa) | Mass Flux (kg/m <sup>2</sup> s) | Quality      |
|--------|-----------------|-------------------|----------------|---------------------------------|--------------|
| 1      | 0.00787–0.016   | 6                 | 14700–14710    | 991–1012                        | 0.395–0.419  |
| 2      | 0.0078–0.011    | 6                 | 9800–9810      | 997–1009                        | 0.512–0.539  |
| 3      | 0.00801–0.00803 | 3–6               | 9800–9810      | 989–1010                        | 0.532–0.594  |
| 4      | 0.00811         | 1–4               | 1980–2020      | 749.9–755.9                     | 0.751–0.777  |
| 5      | 0.008           | 0.99–1            | 6860–17650     | 1993–2019                       | 0.13–0.151   |
| 6      | 0.0134          | 3.658             | 1701–19451     | 2026.4–2068.8                   | 0.366–0.391  |
| 7      | 0.008           | 1.57              | 12,750         | 1579–4973                       | 0.128–0.159  |
| 8      | 0.01            | 4.966             | 16,000         | 1785–6578                       | 0.335–0.353  |
| 9      | 0.00805–0.0085  | 1.8–2             | 9800–9810      | 1455.6–1672.2                   | 0.336–0.437  |
| 10     | 0.008           | 0.99–1            | 17,650         | 1997–2014                       | −0.445–0.234 |

**Fig. 8.** Effect of diameter on CHF by BNN & LUT with experimental data for slice 1 (left) and slice 2 (right).

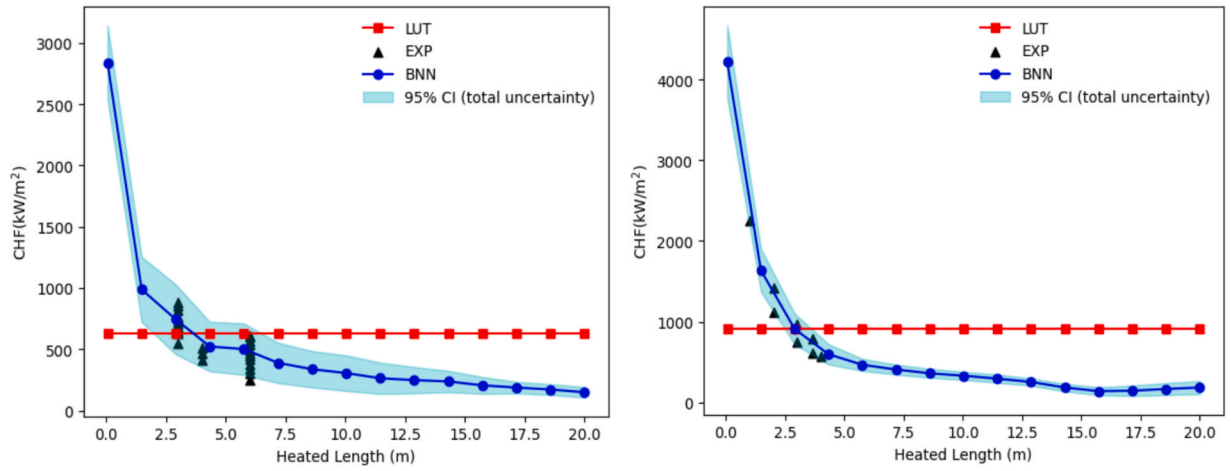


Fig. 9. Effect of heated length on CHF by BNN & LUT with experimental data for slice 3 (left) and slice 4 (right).

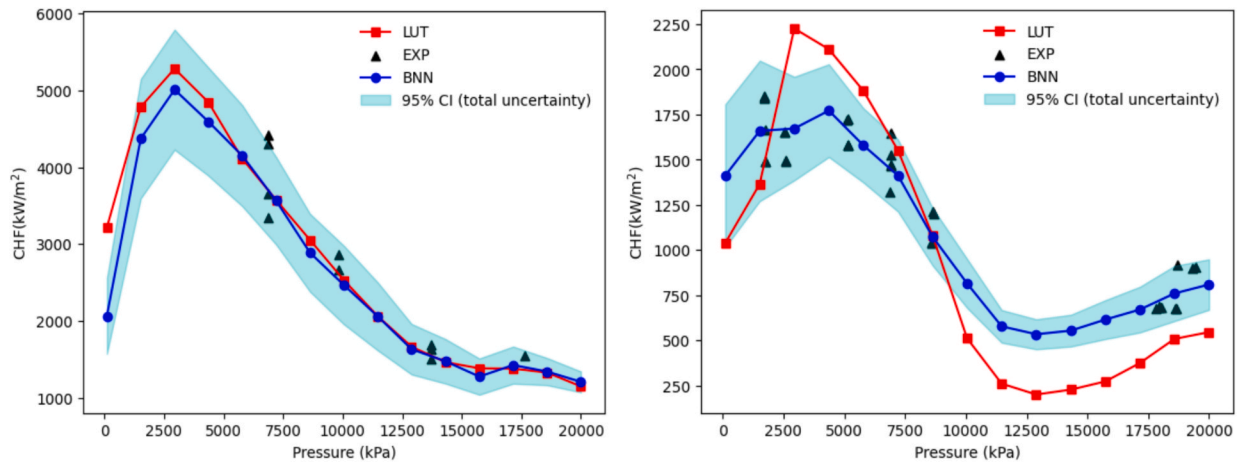


Fig. 10. Effect of pressure on CHF by BNN & LUT with experimental data for slice 5 (left) and slice 6 (right).

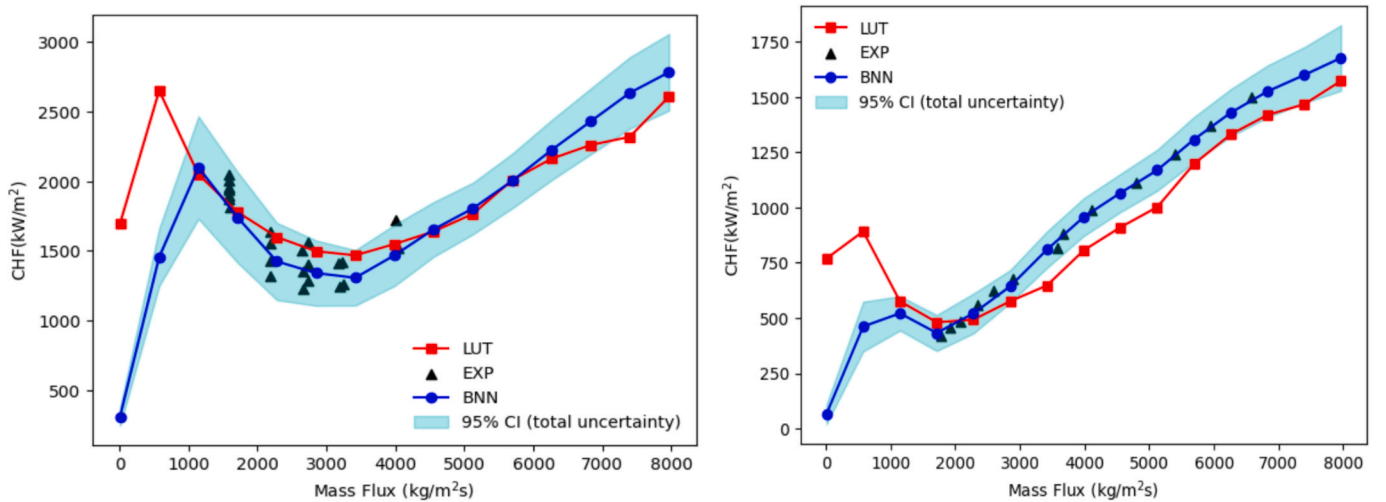


Fig. 11. Effect of mass flux on CHF by BNN & LUT with experimental data for slice 7 (left) and slice 8 (right).

%, and 0.123. Moreover, the  $R^2$  value is also close to 1 for BNN. A notable improvement is also obvious for BNN when the whole data point (24579) is considered for performance evaluation compared to LUT prediction. RMSPE, MAPE, and std P/M have decreased by more than 3 times in the case of BNN. Overall, the CHF prediction accuracy has

improved remarkably after using BNN than LUT. Now, an uncertainty quantification of the BNN model on the test dataset with 95 % CI is shown in Fig. 6. Uncertainty quantification includes plots for epistemic uncertainty, aleatoric uncertainty, and total uncertainty. Fig. 6 shows that aleatoric uncertainty is higher than epistemic uncertainty meaning

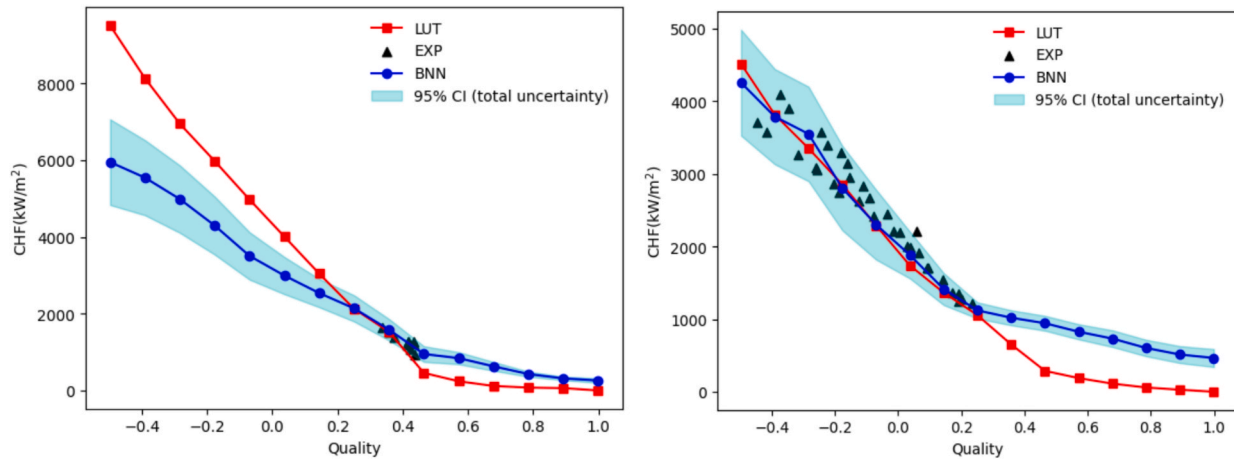


Fig. 12. Effect of quality on CHF by BNN & LUT with experimental data for slice 9 (left) and slice 10 (right).

Table 6

Performance evaluation on slices (experimental data).

|          | Models | Mean P/<br>M | Std P/<br>M | RMSPE<br>(%) | MAPE<br>(%) | R <sup>2</sup> |
|----------|--------|--------------|-------------|--------------|-------------|----------------|
| Slice 1  | BNN    | 0.98         | 0.15        | 15.6         | 12.5        | 0.42           |
|          | LUT    | 1.24         | 0.23        | 34.03        | 25.6        | −0.73          |
| Slice 2  | BNN    | 0.94         | 0.21        | 22.1         | 19.3        | 0.58           |
|          | LUT    | 2.34         | 0.96        | 165.7        | 134.8       | −10.71         |
| Slice 3  | BNN    | 1.07         | 0.18        | 20           | 14.9        | 0.71           |
|          | LUT    | 1.48         | 0.707       | 85.4         | 51.0        | −1.6           |
| Slice 4  | BNN    | 1.01         | 0.11        | 11.5         | 11.1        | 0.96           |
|          | LUT    | 1.01         | 0.379       | 38           | 34          | −0.06          |
| Slice 5  | BNN    | 0.95         | 0.076       | 9.2          | 7.7         | 0.897          |
|          | LUT    | 0.94         | 0.081       | 9.7          | 8.2         | 0.89           |
| Slice 6  | BNN    | 0.99         | 0.07        | 7.2          | 6.0         | 0.94           |
|          | LUT    | 0.95         | 0.21        | 22.1         | 19          | 0.57           |
| Slice 7  | BNN    | 0.96         | 0.035       | 5.3          | 4.5         | 0.87           |
|          | LUT    | 1.02         | 0.076       | 8.1          | 6.9         | 0.77           |
| Slice 8  | BNN    | 1.00         | 0.019       | 1.98         | 1.32        | 0.998          |
|          | LUT    | 0.9          | 0.079       | 12.1         | 10.8        | 0.89           |
| Slice 9  | BNN    | 0.97         | 0.079       | 8.4          | 8           | 0.73           |
|          | LUT    | 0.78         | 0.159       | 26.6         | 21.3        | −1.83          |
| Slice 10 | BNN    | 0.96         | 0.071       | 8.2          | 7.2         | 0.92           |
|          | LUT    | 0.935        | 0.08        | 10.5         | 9.5         | 0.8            |

Table 7

Unseen dataset used for evaluating the BNN model.

|     | Diameter<br>(m) | Heated<br>Length<br>(m) | Pressure<br>(kPa) | Mass<br>Flux<br>(kg/<br>m <sup>2</sup> s) | Quality | CHF (kW/<br>m <sup>2</sup> ) |
|-----|-----------------|-------------------------|-------------------|-------------------------------------------|---------|------------------------------|
| Max | 0.0111          | 1.65                    | 20,000            | 7966                                      | 0.973   | 14763.06                     |
| Min | 0.003           | 0.076                   | 3447              | 39.34                                     | −0.485  | 227.12                       |

most of the uncertainty in the model is due to inherent randomness or variability in the data. For the case of epistemic uncertainty, 43.7 % of actual values (measured CHF) are within 95 % CI. On the other hand, 94.1 % of actual values are within 95 % CI if aleatoric uncertainty is considered. Overall, 94.8 % of actual values are within 95 % CI after measuring total uncertainty that includes both epistemic and aleatoric uncertainty. For a well-calibrated model, around 95 % of measured values should be in 95 % CI. As the fraction of data within 95 % CI is 0.948, it can be considered that the model is capturing uncertainty reasonably well for the test dataset.

To find the prediction accuracy (overprediction or underprediction) of the BNN model in any particular range of input parameters, plots

showing predicted and measured CHF ratios (CHF P/M) against 5 input parameters with  $\pm 30$  % error are illustrated in Fig. 7. Overall, the BNN model didn't show an underprediction greater than 30 % and the majority of data points fell within  $\pm 30$  % range. However, an overprediction of more than 30 % is noticeable for all the parameters. The model shows a higher accuracy (the ratio of predicted and measured CHF is close to 1) in prediction for a mass flux greater than 2000 kg/m<sup>2</sup>s. Maximum overprediction is identified for a quality between 0.4 and 0.7. However, no particular region of overprediction has been identified for pressure and heated length. A few number of points are outside of the  $\pm 30$  % range for all ranges of pressure and heated length except for 20 m. Finally, the model represents higher overprediction for several values of diameters, especially for a tube diameter of around 8 mm. Overall, 96.74 % of the test data fall within the  $\pm 30$  % range for BNN, whereas this value is 81.04 % when LUT is used to predict the same test data.

### 3.2. Prediction on sliced data

To evaluate the prediction performance and physical behavior of the developed model, predictions are made on the slice dataset. For the current study, the slice method and slice dataset have been adopted from the benchmark study described in [24]. Here, in each slice one of the variables is kept as a variable with all others as constant. The varying ranges (varying variables) are divided into 15 equidistant values. The slices described in the OECD/NEA report [24] are shown in Table 4. However, the experimental CHF values at the perfectly sliced data points are not straightforward. So, a short-range in each variable (presented in Table 5) is considered for each sliced data point to get the experimental CHF values from the NRC database. The sliced dataset is formed in a way that captures the effect of each variable (diameter, heated length, pressure, mass flux, and quality) on the CHF. Two slices are considered for each variable (a total of 10 slices for 5 variables). A prediction of perfectly sliced values is made by the developed BNN model and LUT, along with the slices captured from the CHF database (Table 5) shown in Figs. 8–12 with uncertainty band for BNN prediction. The effect of each variable on the CHF is discussed below sequentially considering the prediction from the BNN model and LUT.

As the slices in Table 5 are from the NRC database and experimental CHF is known, it's worthy to represent the performance of BNN and LUT on these slices in terms of evaluation metrics. The performance of BNN and LUT on 10 slices is listed in Table 6. BNN clearly shows the improved performance over LUT for all slices. Although BNN models displayed higher errors (high std P/M, RMSPE, and MAPE values with low R<sup>2</sup> and a value of mean P/M deviated from 1) for slices 1–3, they are still more accurate than LUT. For the slices consisting of pressure, mass flux, and quality as the variable, BNN represents high accuracy—mean P/M close

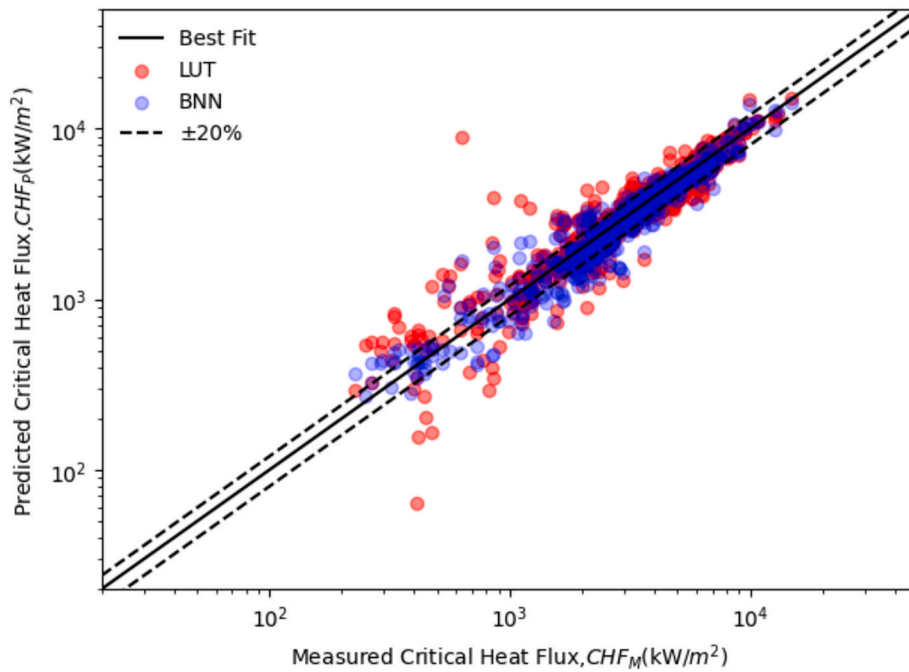


Fig. 13. Comparison of prediction by BNN and LUT on unseen dataset.

Table 8

Performance evaluation of BNN & LUT on the unseen dataset.

| Models | Mean P/M | Std P/M | RMSPE (%) | MAPE (%) | R <sup>2</sup> |
|--------|----------|---------|-----------|----------|----------------|
| BNN    | 0.98     | 0.22    | 22.3      | 14.6     | 0.932          |
| LUT    | 1.07     | 0.637   | 64.05     | 21.6     | 0.87           |

to 1, std P/M less than 0.1, R<sup>2</sup> close to 1, and RMSPE and MAPE less than 10 %.

### 3.2.1. Effect of tube diameter on the CHF (slice 1 and slice 2)

Slices 1 and 2 include the variation of tube diameter while heated length, pressure, mass flux, and quality are kept constant. Fig. 8 demonstrates the variation of CHF with the increase in tube diameter. Each figure includes the prediction of CHF with the change of diameter using BNN and LUT. These plots also include experimental conditions similar to slices 1 and 2. In general, CHFs decrease with the increase in tube diameter while keeping all other local parameters constant. The higher CHF (DNB type) in smaller tubes is due to the larger radial velocity gradient, higher temperature gradient, and reduced distance between the bubbly boundary layer and subcooled core region [47]. On the other hand, a thin liquid film and smooth gas–liquid interface (annular flow region) reduce the probability of entrainment of liquid from the interface to the gas core in a small tube resulting in higher CHF. A steady decrease in CHF with the increase in diameter was observed for LUT for both slices. However, in the case of BNN, CHF decreased initially with an increase in tube diameter (up to around 8 mm) and later showed an increasing trend before decreasing again for the higher tube diameter. The experimental data points supported this fluctuating trend in CHF observed by BNN. The transition in CHF after 8 mm might be due to a change in flow dynamics or flow regimes [48].

### 3.2.2. Effect of tube heated length on the CHF (slice 3 and slice 4)

A comparison of the effect of heated length on CHF is illustrated in Fig. 9 for slices 3 and 4 with constant tube diameter, mass flux, pressure, and outlet quality. It is concluded in the literature that the effect of heating length on CHF disappears in subcooled boiling for a tube length of more than 200 mm [49]. Some studies showed a linear decrease [50],

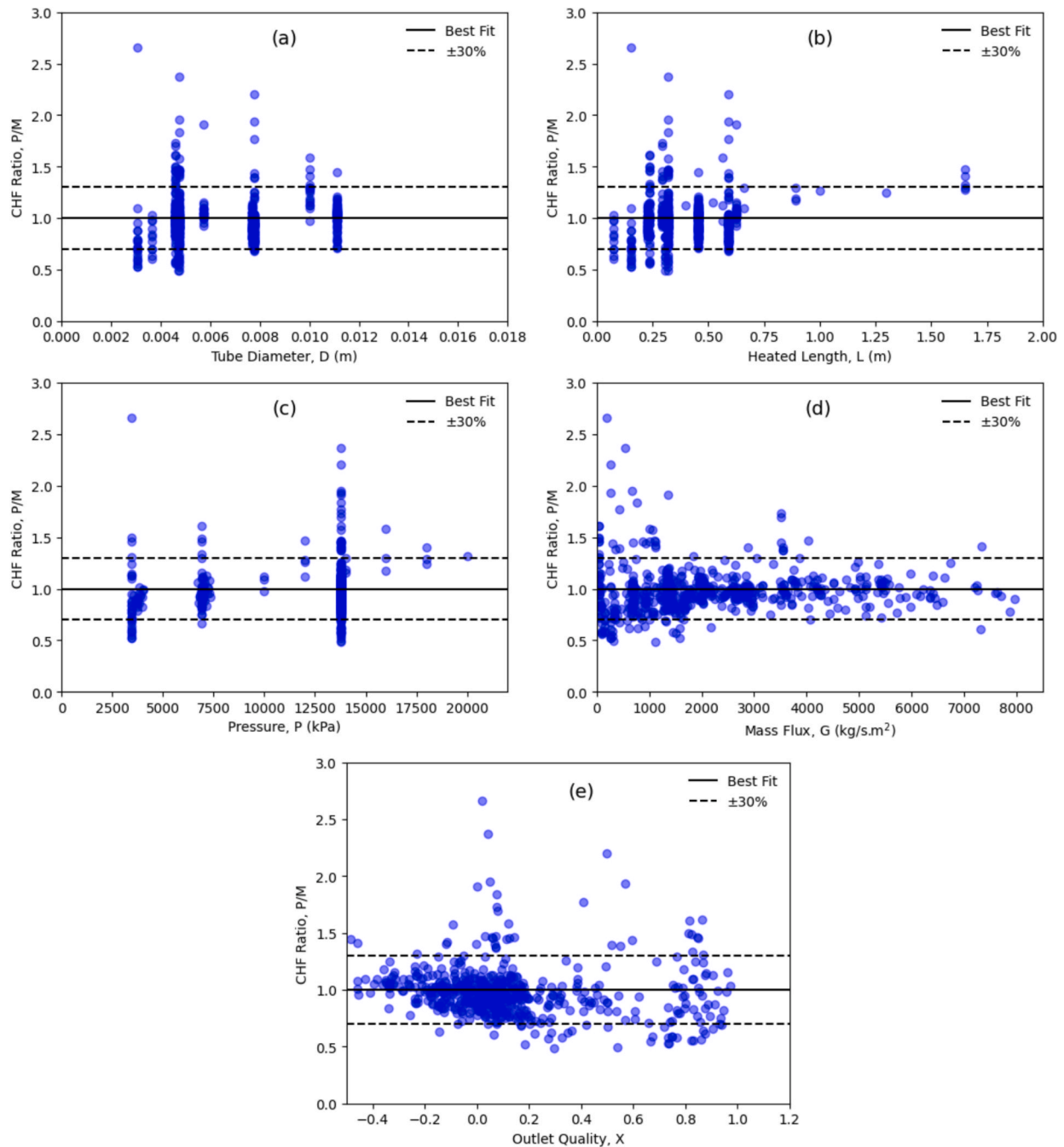
and a monotonic decrease [51] of CHF with the increase of heated length. It was also found that the effect of heated length is profound in shorter tubes. For the present case, BNN represented a decreasing trend with the increase of heated length for both slices. A sharp decreasing trend in CHF was visible for the shorter heated length. Experimental data also followed the same trend of BNN, especially for slice 4. However, LUT predicted a horizontal line of constant values for both slices as a prediction from the LUT doesn't depend on heated length.

### 3.2.3. Effect of pressure on the CHF (slice 5 and slice 6)

The effect of pressure on the CHF for slices 5 and 6 is shown in Fig. 10. The increase in pressure showed an increasing and decreasing trend for both slices. The values and trend for slice 5 perfectly matched BNN, LUT, and experimental data points. Although the prediction for slice 6 from BNN and LUT represented the same trend, LUT over-predicted for low pressure (3 MPa–7 MPa) and underpredicted for higher pressure (between 10 MPa–21 MPa) compared to BNN and experimental data points. However, BNN followed the experimental trend exactly and matched with values reasonably well. The trend of increasing CHF with the increase of pressure at a low range and decreasing CHF at a high-pressure range as critical pressure is approached has been confirmed by experimental studies [49,52].

### 3.2.4. Effect of mass flux on the CHF (slice 7 and slice 8)

The effect of mass flux on the CHF for slices 7 and 8 is presented in Fig. 11. The BNN, LUT, and experimental data points demonstrated the same pattern of variation of CHF with mass flux. CHF initially increased with the increase of mass flux, then decreased with the increase of mass flux, and later increased again with a further increase of mass flux. Note that the fluctuation (initial increase and then decrease) in CHF happened at low mass flux region (less than 3000 kg/m<sup>2</sup>s for slice 7 and less than 2000 kg/m<sup>2</sup>s for slice 8) for slices. CHF at low mass flux is mainly dominated by local conditions in subcooled boiling. So, based on heated length (or L/D), a fluctuation in the CHF might be observed at low flow mass flux [53]. However, in high mass flux, CHF is dependent on the global phenomenon (influenced by void fraction in the bulk liquid) and increases consistently with the increase of mass flux. The bubbles on the heated surface are swept away with high mass flux in a subcooled region which prevents the formation of a bubble blanket on the surface and



**Fig. 14.** Prediction accuracy on unseen data for input parameters. a) for diameter. b) for heated length. c) for pressure. d) for mass flux and e) for quality.

results in an increase of CHF. On the other hand, in the high-quality region, the liquid film thickness on the surface becomes thicker with high mass flux [54,55], and the dryout condition gets delayed because of this increased film thickness on the surface. Moreover, with the increase of mass flux, turbulence is also increased in the tube which further promotes bubble detachment from the surface and phase mixing. In both cases heat transfer coefficient is increased which ensures high CHF.

### 3.2.5. Effect of quality on the CHF (slice 9 and slice 10)

The variation of CHF with the increase in exit quality is displayed in Fig. 12 for slices 9 and 10. BNN, LUT, and the experimental data points showed the same trend of changing CHF with the increase in exit quality. CHF decreased with the increase in outlet quality for all cases. This trend of CHF with exit quality is also supported by correlations [15], and experimental works [56]. Both BNN and LUT display a sharp decrease in

CHF in the subcooled to low-quality region and a gradual increase in high-quality to near dryout quality region. In general, high quality indicates higher vapor in the tube outlet which works as a vapor blanket for subcooled flow, and thinner liquid film for annular flow resulting in a decrease of CHF. However, some studies also reported that CHF might increase in the churn, churn-annular, or annular flow near transition due to the stabilization of interfacial waves at increased void in the tube [57].

### 3.3. Prediction of the unseen dataset

The test dataset is a part of the total dataset although not used for training and validation of the developed BNN model. So, the model should have a better prediction on the test dataset. To verify the prediction accuracy of BNN over LUT a further prediction was made on the



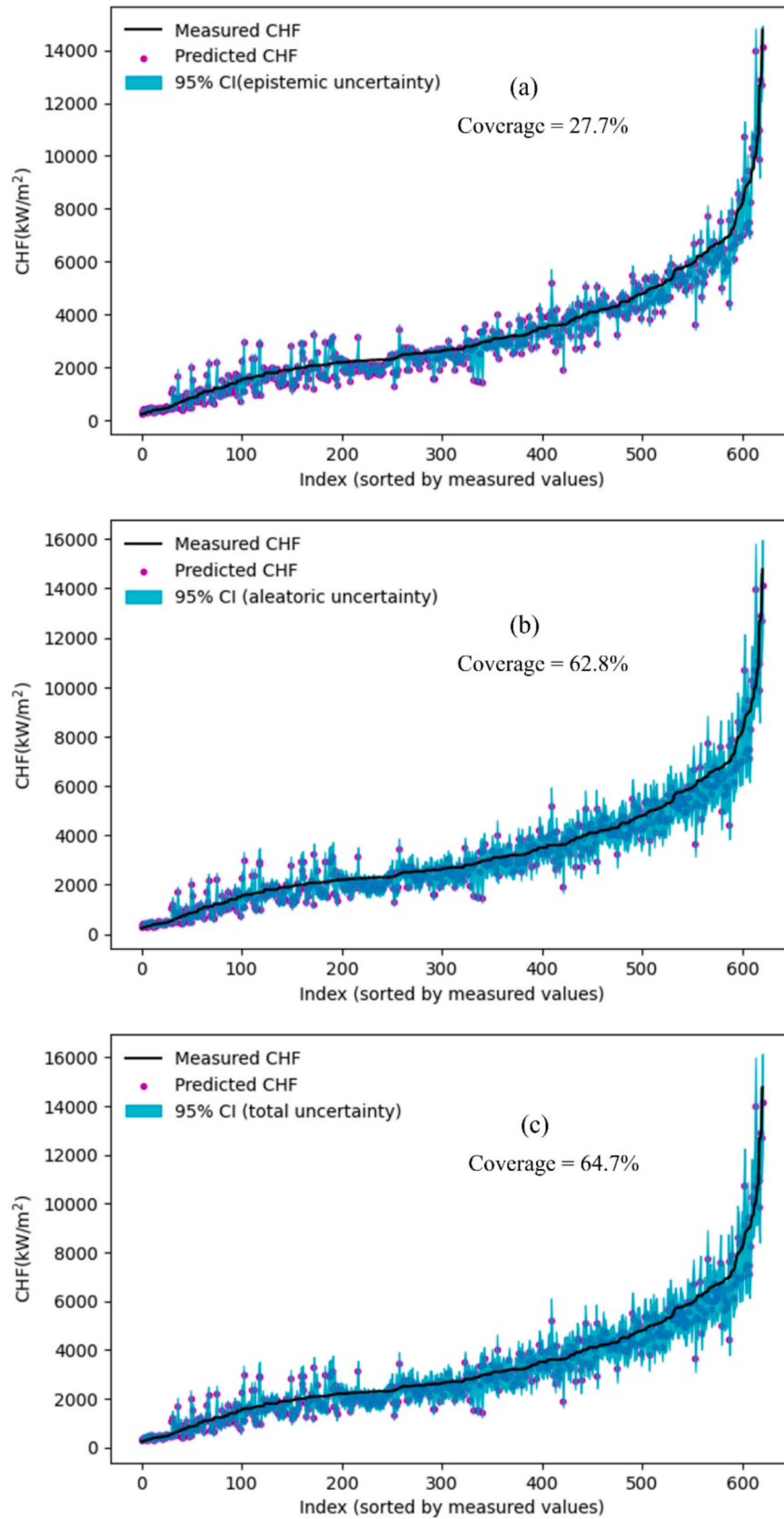


Fig. 15. Uncertainty quantification of unseen data with 95% CI. a) epistemic uncertainty. b) aleatoric uncertainty. c) total uncertainty.

**Table 9**

Extrapolated tube dataset used for performance evaluation.

|     | Diameter (m) | Heated Length (m) | Pressure (kPa) | Mass Flux (kg/m <sup>2</sup> s) | Quality | CHF (kW/m <sup>2</sup> ) |
|-----|--------------|-------------------|----------------|---------------------------------|---------|--------------------------|
| Max | 0.0375       | 1.972             | 18960.6        | 10,864                          | 0.20    | 12901.9                  |
| Min | 0.0019       | 0.152             | 6890           | 666                             | −0.459  | 1900                     |

**Table 10**

Performance evaluation of the BNN model on extrapolated tube dataset.

| Model | Mean P/M | Std P/M | RMSPE (%) | MAPE (%) | R <sup>2</sup> |
|-------|----------|---------|-----------|----------|----------------|
| BNN   | 0.910    | 0.30    | 32.1      | 25.4     | 0.28           |

unseen dataset. The unseen dataset was collected from the online available literature [58–60]. This unseen dataset is not a part of the training or testing dataset. However, these points are in the range of input parameters. The minimum and maximum range of the unseen dataset (a total of 621 data points) is given in Table 7, and the prediction on the unseen dataset by BNN and LUT is shown in Fig. 13 with  $\pm 20\%$  error range. The performance evaluation considering five evaluation metrics is listed in Table 8 for both models. The CHF prediction accuracy for the BNN is higher than LUT. The RMSPE and std P/M for BNN are 2.5 times lower than LUT. Moreover, BNN shows higher R<sup>2</sup> values and the mean P/M is close to 1 compared to LUT. Lastly, BNN predicts unseen datasets with a MAPE of 14.6 % whereas this value is 21.9 % for LUT. Now, the effect of input parameters (diameter, heated length pressure, mass flux, and quality) on prediction accuracy is illustrated in Fig. 14 for the unseen dataset with  $\pm 30\%$  error range. Likewise test dataset, the

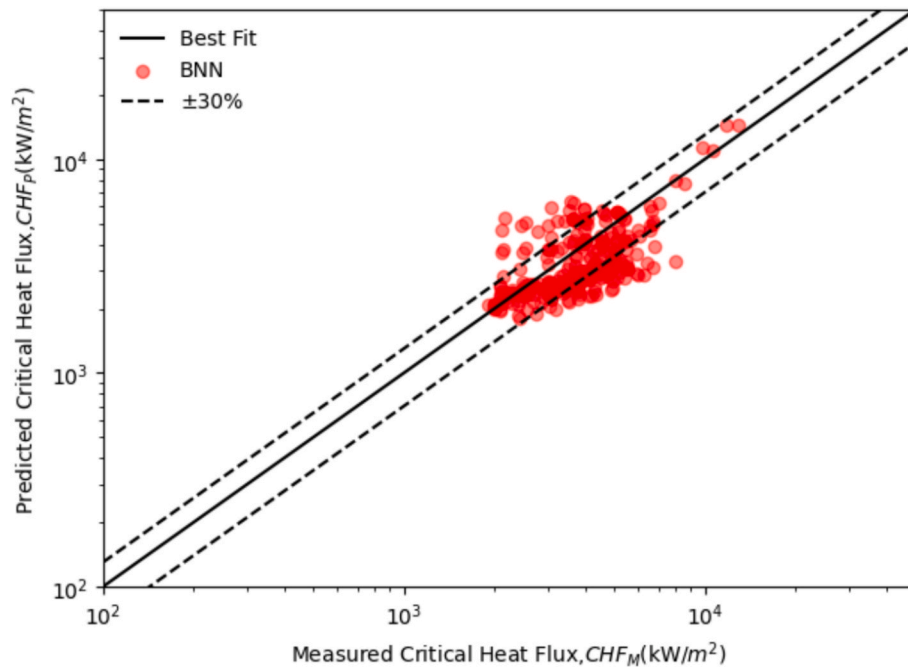
model is more accurate for higher ( $>1000 \text{ kg/m}^2\text{s}$ ) mass flux, almost all the data are within the  $\pm 30\%$  range. Any particular value or range of tube diameter or pressure is not identified where the model shows overprediction or underprediction consistently. A small number of points are outside  $\pm 30\%$  for each diameter or pressure tested here. The effect of heated length on the accuracy shows a repeated underprediction of greater than  $-30\%$  for a heated length of less than 0.457 m. The majority of the data points are within  $\pm 30\%$  when the effect of quality on the CHF for the unseen dataset is shown in Fig. 14(e). Overall, 88.24 % of unseen data points are within  $\pm 30\%$  compared to 83.09 % for LUT.

Fig. 15 presents the uncertainty quantification (epistemic uncertainty, aleatoric uncertainty, and total uncertainty) for the unseen dataset with 95 % CI. Predicted CHFs are more scattered for unseen datasets compared to the CHFs predicted on the test dataset. The prediction intervals for aleatoric uncertainty are wider than prediction intervals for epistemic uncertainty likewise the uncertainties for the test dataset. Here, around 65 % of data is captured within 95 % CI when total uncertainty is considered. In addition, epistemic uncertainty includes 62.8 % of data within 95 % CI, whereas for the aleatoric uncertainty, this fraction is 0.277. This underestimation in uncertainty indicates that the BNN model is not as accurate on the unseen dataset as it was for the test dataset.

**Table 12**

Performance evaluation of the BNN model on the annulus dataset.

| Model | Mean P/M | Std P/M | RMSPE (%) | MAPE (%) | R <sup>2</sup> |
|-------|----------|---------|-----------|----------|----------------|
| BNN   | 1.11     | 0.29    | 32.04     | 23.8     | 0.53           |

**Fig. 16.** Prediction on the extrapolated dataset (tube) by BNN.**Table 11**

Annulus dataset used for model performance evaluation.

|     | Dia. (hydraulic) (m) | Dia. (heated) (m) | Heated Len. (m) | Pressure (kPa) | Mass Flux (kg/m <sup>2</sup> s) | Quality | CHF    |
|-----|----------------------|-------------------|-----------------|----------------|---------------------------------|---------|--------|
| Max | 0.022                | 0.096             | 2.74            | 15513.2        | 6104                            | 0.615   | 5991.1 |
| Min | 0.0045               | 0.0113            | 0.736           | 4088.5         | 196                             | −0.32   | 429.06 |

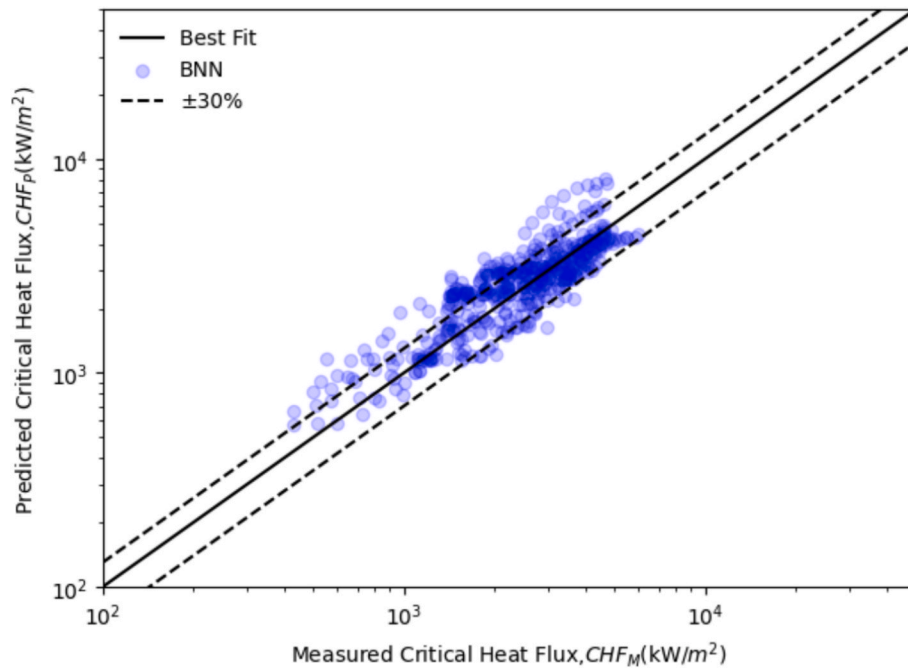


Fig. 17. Prediction on the extrapolated dataset (annulus) by BNN.

Table 13

Performance comparison of BNN model with other ML models.

| Models              | Performance metrics |         |           |          |                |
|---------------------|---------------------|---------|-----------|----------|----------------|
|                     | Mean P/M            | Std P/M | RMSPE (%) | MAPE (%) | R <sup>2</sup> |
| LUT [23]            | 1.032               | 0.361   | 36.3      | 19.77    | 0.941          |
| Benchmark NN [24]   | 1.010               | 0.115   | 12.6      | 8.03     | 0.978          |
| NN [65]             | 1.034               | 0.201   | 20.4      | 11.20    | 0.9802         |
| CNN [65]            | 1.027               | 0.166   | 16.80     | 9.72     | 0.9864         |
| Transformer [65]    | 1.008               | 0.122   | 12.30     | 7.22     | 0.9874         |
| Best Single NN [66] | 1.008               | 0.106   | 10.60     | 6.87     | 0.9854         |
| Best Ensemble [66]  | 1.007               | 0.104   | 10.43     | 6.76     | 0.9854         |
| BNN (Current)       | 1.003               | 0.106   | 10.6      | 6.5      | 0.989          |

### 3.4. Prediction on the extrapolated dataset

All the data points (experimental conditions) used for the above comparison include the data ranges used for training the BNN model. To evaluate the performance of the model on the extrapolated data points (a range not considered for training the model), a further comparison is shown for extrapolated data points. These extrapolated data points include the geometry (diameter, flow configuration), and operating conditions (mass flux) outside of the training range. The dataset was taken from the online available source. Here two types of extrapolated conditions have been considered. First, the diameter of the uniformly heated tube is out of the range of 2–16 mm (1.9 mm or 22.2–37.5 mm) or mass flux exceeds 8000 kg/m<sup>2</sup>s [61,62]; second, the annulus geometry with uniform heat flux [63,64] that was not considered to train the BNN model. The extrapolated tube diameter and mass flux (a total of 275 data points) with range are shown in Table 9. As LUT is not applicable for this diameter range, performance evaluation by BNN is depicted in Table 10. BNN has a higher std P/M, RMSPE, and MAPE on extrapolated data points compared to training, test, or in-range datasets. The BNN model exhibits mean P/M, std P/M, RMSPE, MAPE, and R<sup>2</sup> of 0.91, 0.3, 32.1 %, 25.4 %, and 0.28 on the extrapolated data points respectively, with 67.6 % of total data points within  $\pm 30$  % range. Moreover, a comparison

between measured CHF and predicted CHF by BNN is illustrated in Fig. 16 with  $\pm 30$  % error range.

Furthermore, a prediction was made on the annulus tube to check the performance of the developed BNN on different geometry configurations. The range of the annulus dataset (a total of 489 data points) is shown in Table 11. In the case of annulus geometry, two types of diameters are usually considered—hydraulic diameter and heated diameter. For the current evaluation, both types of diameters were tested, and a relatively small error was obtained after using hydraulic diameter as input, and the performance of BNN in terms of evaluation metrics is shown in Table 12. The annulus dataset shows mean P/M, std P/M, RMSPE, MAPE, and R<sup>2</sup> of 1.11, 0.29, 32.04 %, 23.8 %, and 0.53 respectively, with around 70 % of total data points within  $\pm 30$  % range. A comparison of experimental and BNN prediction on the annulus dataset is illustrated in Fig. 17 with  $\pm 30$  %. The overall finding of the BNN model on the extrapolated dataset indicates that the current model is less accurate on extrapolated datasets compared to in-range datasets.

### 3.5. Performance comparison with other models trained on the same OECD dataset

In this study, a Bayesian Neural Network (BNN) model has been developed using the OECD/NRC CHF database. While several other Machine Learning (ML) models have also been developed based on the same database [65,66], these include neural networks, transfer learning approaches, ensembles of neural networks, convolutional neural networks, and others. Although the dataset is identical, variations in random data splitting result in different training and testing sets across studies. To enable a fair comparison, the performance of the proposed BNN model has been evaluated on the entire dataset comprising 24,579 data points. Table 13 presents a comparative analysis of the BNN model against other ML models using five evaluation metrics. The BNN model achieved the lowest mean P/M and MAPE values, along with the highest R<sup>2</sup>, indicating superior overall performance. Although the ensemble neural network model recorded the lowest standard deviation of P/M (0.104) and RMSPE (10.43 %), the BNN model closely followed with a std P/M of 0.106 and RMSPE of 10.6 %. It is important to note that BNN is a probabilistic ML model, whereas all other models used in the comparison are deterministic. Deterministic models generally converge

more easily, as they predict only a single output (CHF). In contrast, the BNN model predicts both the CHF and its associated uncertainty. Despite this additional complexity, the BNN demonstrates competitive performance compared to other ML models.

#### 4. Conclusion

In this study, a Bayesian Neural Network (BNN) for the prediction of CHF with both epistemic and aleatoric uncertainties has been developed using the NRC CHF database considering the outlet condition. The performance of the developed model has been compared with the widely used LUT in terms of five evaluation metrics— mean P/M, std P/M, RMSPE, MAPE, and  $R^2$ . The performance evaluation was made on the prediction on test, unseen, and extrapolated datasets. The finding of the present study is summarized as follows:

1. The developed BNN model with tube diameter, heated length, pressure, mass flux, and quality as inputs exhibit a better performance of predicting CHF than widely used LUT, although the same dataset has been utilized for developing both BNN and LUT. The BNN model outperforms LUT in terms of each evaluation metric considered here for performance evaluation on the test, whole NRC database, and unseen dataset.
2. The BNN model predicts the test and the whole NRC database with a mean P/M, std. P/M, RMSPE, MAPE, and  $R^2$  of 1.007, 0.123, 12.3 %, 7.6 %, 0.986, and 1.003, 0.106, 10.6 %, 6.5 %, 0.989 respectively, whereas these values are 1.029, 0.328, 32.9 %, 19.8 %, 0.941, and 1.032, 0.361, 36.3 %, 19.77 % and 0.941 for LUT.
3. The BNN model captures around 94.8 % of the test dataset in 95 % CI which is an indication that it's capable of predicting mean CHF as well as uncertainty associated with the prediction reasonably well. The models also display both epistemic and aleatoric uncertainties separately for test and unseen datasets. Although it captures 94.8 % of test data in 95 % CI, it includes only 64.8 % unseen data in 95 % CI.
4. The trend and values of CHF for the slice datasets after using BNN confirm that the developed model is capturing the physics of the variation of CHF with tube diameter, heated length, pressure, mass flux, and outlet quality. The trends by BNN for slice data agree with the experimental findings.
5. The BNN model predicted the Critical Heat Flux (CHF) for the extrapolated tube dataset with RMSPE and MAPE values of 32.1 % and 25.4 %, respectively. For the extrapolated annular tube geometry, the corresponding values were 32.04 % and 23.8 %. Although the model exhibited lower accuracy compared to its performance on the in-range dataset, it may still be applicable for CHF prediction on extrapolated datasets.
6. The developed model captures uncertainty for the test dataset. However, the model didn't capture the uncertainty for the unseen dataset. So, further tuning of the model parameters might be required for better accuracy in uncertainty prediction. On the other hand, other machine learning algorithms that capture both mean prediction and uncertainty can be considered for better accuracy. Furthermore, physics physics-informed neural network (PINN) that predicts mean value as well as captures inherent physics can also be utilized.

#### CRedit authorship contribution statement

**Anadi Mondal:** Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Subash L. Sharma:** Writing – review & editing, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Data availability

The data that has been used is confidential.

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